Pneumatic Sample-Lift Mechanism

Use a pneumatic actuator to lift or position a sample while controlling motion and air use.

Intermediate	1-2 class periods	Skill Builder

Engineering Context

A rover must transfer a soil sample from the ground to an onboard analysis tray. Your team will prototype the actuator portion of the mechanism.

What You Will Practice

- Describe pneumatic energy conversion
- Select cylinder motion and linkage
- Control extend/retract sequence
- Identify pinch and pressure hazards

What You Need

- Low-pressure classroom pneumatic kit
- Cylinder
- Tubing and fittings
- Control valve or programmed solenoid
- Lightweight sample object
- Engineering notebook

Student Instructions

1. Define the start and end positions of the sample.
2. Sketch two mechanisms using the cylinder directly or through a linkage.
3. Select one concept and identify pinch points.
4. Assemble the circuit with pressure off.
5. Check tubing and fittings before enabling air.
6. Run manual low-pressure extension and retraction.
7. Add control logic or a repeatable operating procedure.
8. Complete ten transfer cycles and record faults or air leaks.

Engineering Requirements

- Move the sample through the required height or angle
- No uncontrolled dropping
- Complete ten cycles without a tubing disconnect
- Include a safe default state
- Document the pneumatic diagram

Constraints & Safety

- Depressurize before changing tubing
- Keep fingers clear of pinch points
- Use only instructor-set pressure
- Wear eye protection if required

What You Submit

- Concept sketches
- Pneumatic diagram
- Final mechanism photo
- Cycle test log
- Safety analysis

Reflection Questions

1. Why was the cylinder bore or stroke appropriate?
2. Where could the mechanism jam?
3. What happened when air flow changed?
4. How could a sensor confirm motion?

Engineering Tip Arrange the mechanism so a loss of pressure moves the system toward a safe state whenever possible.	**Common Mistake** Changing tubing while the system is pressurized can cause sudden motion or hose release.
Stretch Goal Add a sensor interlock that prevents the next step until the cylinder reaches position.	**Career Connection** Fluid-power and mechatronics engineers use pneumatic systems for clamps, deployment devices, and factory support equipment.